

"Hot and Cold Radar"

The concept of this project is a Hot and Cold-like game that you can technically play with two players, or you can play alone. I decided to give it a radar-like appearance as well, with a green technology aesthetic.

The way you use it is with the potentiometers and button if you are controlling the radar, and the arrow keys if you are controlling the target. One potentiometer controls the x position of the radar and the other controls the y axis, indicated by paper signs that tell you as such. Meanwhile, you press the button when you think you are on the target.

The red light indicates you miss, the green light indicates that you hit it and are temporarily stopped (the red light that follows indicates you can move again), and the white light's brightness increases the closer you are to the target. When it's at its brightest, you are on the target and can press the button. Meanwhile, the target moves very slowly but if it reaches the edge of the canvas, it will be teleported to a random point of the map, indicated by a unique sound from the buzzer to let the radar controlling know the target's position changes drastically.

It works by detecting if the radar's x and y location is within the height and width of the target, as well as if its location is within the two warning layers. When it is in one of these areas, it will send out 'C' when it isn't within any warning borders, 'D' when it's within the furthest border, 'E' when it's within the closest border, and 'F' if it's on the target. Each of these messages are sent back to the breadboard, indicating how bright the white light should shine, with C not shining at all and F shining the brightest.

This messaging system is also used to shine the green and red lights, as well as sound the indicator that the target teleported to a random location. Message 'A' will temporarily shine the green light, before switching to red and playing a buzzing sound to indicate the radar player they can move again. Message 'B' will only shine the red light and do the buzzing sound. Both of these messages will only play when the button sensor is detected, meanwhile Message 'G' will use no lights, only doing a set of buzzer frequencies.

The program detects whether or not the target's x or y location has either gone below zero, or has gone past the width or height of the canvas, using `random()` to determine its new location. `Random()` is also used to change the target position if it is successfully found.

Photos:

